

12.2 Confidence ellipsoids

Two 95% intervals, one for each of two coefficients, do not make a 95% statement about the pair, and they ignore the correlation between the estimates. The natural joint region is an ellipsoid: the set of values the F test does not reject, shaped like the covariance matrix of the estimates, with shadows that are wider than the t intervals. This geometry explains an F test that rejects while every t test accepts, and the reverse.

12.2.1 The ellipsoid

Let $\mathbf{\Lambda}$ be a $p \times q$ matrix of rank q whose columns lie in $\mathcal{C}(\mathbf{X}^\top)$, so that every entry of $\boldsymbol{\phi} = \mathbf{\Lambda}^\top \boldsymbol{\beta}$ is estimable. Put

$$\hat{\boldsymbol{\phi}} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}, \quad \mathbf{W} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}.$$

By Theorem 9.2.4(a), neither depends on the choices of $\hat{\boldsymbol{\beta}}$ and $\mathbf{G}$, and $\mathbf{W}$ is positive definite. By Theorem 7.5.1(a), $\hat{\boldsymbol{\phi}} \sim N_q(\boldsymbol{\phi}, \sigma^2 \mathbf{W})$.

Theorem 12.2.1 — Confidence ellipsoid for $\mathbf{\Lambda}^\top \boldsymbol{\beta}$

With the notation above, let $F_\alpha = F_\alpha(q, n - r)$ and

$$E(\mathbf{Y}) = \{\boldsymbol{\phi}_0 \in \mathbb{R}^q : (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}_0)^\top \mathbf{W}^{-1} (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}_0) \leq q s^2 F_\alpha\}. \quad (12.2.1)$$

- (a) $E(\mathbf{Y})$ is an exact $1 - \alpha$ confidence set for $\mathbf{\Lambda}^\top \boldsymbol{\beta}$.
- (b) $E(\mathbf{Y})$ is a solid ellipsoid centred at $\hat{\boldsymbol{\phi}}$. If $\mathbf{W} = \sum_{k=1}^q w_k \mathbf{u}_k \mathbf{u}_k^\top$ is a spectral decomposition, its principal axes point along $\mathbf{u}_1, \dots, \mathbf{u}_q$ and have half-lengths $\sqrt{q s^2 F_\alpha w_k}$. Its volume is

$$\frac{\pi^{q/2}}{\Gamma(q/2 + 1)} (q s^2 F_\alpha)^{q/2} \det(\mathbf{W})^{1/2}.$$

- (c) (*Duality with the F test.*) For $\mathbf{d} \in \mathbb{R}^q$, let $\text{SSE}_{\mathbf{d}}$ be the minimum of $\|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2$ over $\{\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}\}$ and let

$$F_{\mathbf{d}} = \frac{(\text{SSE}_{\mathbf{d}} - \text{SSE})/q}{s^2} = \frac{(\hat{\boldsymbol{\phi}} - \mathbf{d})^\top \mathbf{W}^{-1} (\hat{\boldsymbol{\phi}} - \mathbf{d})}{q s^2}.$$

Then $\mathbf{d} \in E(\mathbf{Y})$ iff $F_{\mathbf{d}} \leq F_\alpha$, that is, iff the level- α F test of $H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ does not reject.

Proof.

- (a) Since $\mathbf{W}$ is invertible, $\mathbf{W}^{-1}$ is its only generalized inverse, and Corollary 7.5.3 with $\text{rank}(\mathbf{\Lambda}) = q$ gives

$$\frac{(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})^\top \mathbf{W}^{-1} (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})}{q s^2} \sim F(q, n - r)$$

at every parameter value. This is a pivot, and $\boldsymbol{\phi} \in E(\mathbf{Y})$ is the event that it is at

most F_α .

- (b) Put $c = q s^2 F_\alpha$ and write $\phi_0 - \hat{\phi} = \sum_k c_k \mathbf{u}_k$. Since $\mathbf{W}^{-1} = \sum_k w_k^{-1} \mathbf{u}_k \mathbf{u}_k^\top$, the defining inequality becomes $\sum_k c_k^2 / w_k \leq c$, the standard form of an ellipsoid with half-axes $\sqrt{c w_k}$ along $\mathbf{u}_k$. Equivalently $E(\mathbf{Y}) = \hat{\phi} + \mathbf{W}^{1/2} B$, where B is the ball of radius $\sqrt{c}$ in $\mathbb{R}^q$. A linear map multiplies volume by the absolute value of its determinant, so the volume is $\det(\mathbf{W})^{1/2} c^{q/2}$ times the volume $\pi^{q/2} / \Gamma(q/2 + 1)$ of the unit ball.
- (c) The two expressions for $F_{\mathbf{d}}$ agree by Theorem 9.2.4(c), and the equivalence is the definition of $E(\mathbf{Y})$.

■

The statistic $F_{\mathbf{d}}$ is the F statistic of the general linear hypothesis in both of its forms (see Theorem 11.3.2), and part (c) is Proposition 12.1.2 in concrete form. In particular the test of $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{0}$ rejects iff the ellipsoid misses the origin.

In the full-rank case with $\mathbf{A} = \mathbf{I}_p$, the region for the whole coefficient vector is

$$\begin{aligned} \{\mathbf{b} : (\hat{\boldsymbol{\beta}} - \mathbf{b})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \mathbf{b}) &\leq p s^2 F_\alpha(p, n - p)\} \\ &= \{\mathbf{b} : \|\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}\mathbf{b}\|^2 \leq p s^2 F_\alpha(p, n - p)\}. \end{aligned}$$

The second form says that $\mathbf{b}$ is in the region iff its fitted vector $\mathbf{X}\mathbf{b}$ is close to the least squares fit $\hat{\mathbf{Y}}$. This suggests looking at the mean vector directly.

12.2.2 A ball in the column space

Proposition 12.2.2 — Confidence ball for the mean vector

Let $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ and $c_r = r s^2 F_\alpha(r, n - r)$. The set

$$B(\mathbf{Y}) = \{\boldsymbol{\mu}_0 \in \mathcal{C}(\mathbf{X}) : \|\hat{\mathbf{Y}} - \boldsymbol{\mu}_0\|^2 \leq c_r\}$$

is an exact $1 - \alpha$ confidence set for $\boldsymbol{\mu}$: a ball of radius $\sqrt{c_r}$ in the r -dimensional space $\mathcal{C}(\mathbf{X})$, centred at the fitted vector. Moreover, if $\mathbf{A} = \mathbf{X}^\top \mathbf{T}$ for an $n \times q$ matrix $\mathbf{T}$ (such a $\mathbf{T}$ exists by Theorem 8.3.1), the image of $B(\mathbf{Y})$ under $\boldsymbol{\mu}_0 \mapsto \mathbf{T}^\top \boldsymbol{\mu}_0$ is the set

$$\{\phi_0 : (\hat{\phi} - \phi_0)^\top \mathbf{W}^{-1} (\hat{\phi} - \phi_0) \leq c_r\},$$

which contains the ellipsoid [Eq. 12.2.1](#).

Proof. $\hat{\mathbf{Y}} - \boldsymbol{\mu} = \mathbf{M}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$, and $\mathbf{M}$ is symmetric idempotent of rank r , so $\|\hat{\mathbf{Y}} - \boldsymbol{\mu}\|^2 / \sigma^2 \sim \chi^2(r)$ by Theorem 4.3.2(a), independently of SSE (Theorem 7.5.1(c)). Dividing by $r s^2 / \sigma^2$ gives an $F(r, n - r)$ pivot, and the first claim follows as before.

For the image, put $\mathbf{A} = \mathbf{T}^\top \mathbf{M}$, a $q \times n$ matrix. For $\boldsymbol{\mu}_0 \in \mathcal{C}(\mathbf{X})$, $\mathbf{T}^\top \boldsymbol{\mu}_0 = \mathbf{A}\boldsymbol{\mu}_0$, and $\mathbf{A}\mathbf{A}^\top = \mathbf{T}^\top \mathbf{M}\mathbf{T} = \mathbf{W}$ (Theorem 9.2.4(a)). Also $\hat{\phi} = \mathbf{T}^\top \hat{\mathbf{Y}} = \mathbf{A}\hat{\mathbf{Y}}$. So the image consists of the points $\hat{\phi} - \mathbf{A}\mathbf{u}$ with $\mathbf{u} \in \mathcal{C}(\mathbf{X})$ and $\|\mathbf{u}\|^2 \leq c_r$. If $\mathbf{z} = \mathbf{A}\mathbf{u}$ with $\|\mathbf{u}\|^2 \leq c_r$, then $\mathbf{z}^\top \mathbf{W}^{-1} \mathbf{z} = \mathbf{u}^\top \mathbf{A}^\top \mathbf{W}^{-1} \mathbf{A} \mathbf{u} \leq \|\mathbf{u}\|^2 \leq c_r$, because $\mathbf{A}^\top \mathbf{W}^{-1} \mathbf{A}$ is symmetric and idempotent, hence a projection, and a projection does not increase length (Proposition 6.3.4(d)). Conversely, if $\mathbf{z}^\top \mathbf{W}^{-1} \mathbf{z} \leq c_r$, then $\mathbf{u} = \mathbf{A}^\top \mathbf{W}^{-1} \mathbf{z}$ lies in

$\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$, satisfies $\mathbf{A}\mathbf{u} = \mathbf{z}$, and has $\|\mathbf{u}\|^2 = \mathbf{z}^\top \mathbf{W}^{-1} \mathbf{z} \leq c_r$.

For the containment it remains to show $qF_\alpha(q, \nu) \leq rF_\alpha(r, \nu)$ when $q \leq r$. Let $U_q \sim \chi^2(q)$, $U' \sim \chi^2(r-q)$ and $V \sim \chi^2(\nu)$ be independent. Then $U_q + U' \sim \chi^2(r)$, so for every c , $\Pr\{(U_q + U')/(V/\nu) \leq c\} \leq \Pr\{U_q/(V/\nu) \leq c\}$. The upper α point of the variable on the left is $rF_\alpha(r, \nu)$, and that of the one on the right is $qF_\alpha(q, \nu)$. ■

So all confidence ellipsoids are views of one ball around $\hat{\mathbf{Y}}$ in the column space. Each linear summary $\mathbf{\Lambda}^\top \boldsymbol{\beta}$ maps the ball to an ellipsoid, and since the ball covers $\boldsymbol{\mu}$ with probability $1 - \alpha$, all the images cover their targets *simultaneously* with probability at least $1 - \alpha$, at the price of the constant $rF_\alpha(r, \nu)$ in place of $qF_\alpha(q, \nu)$. Section 12.4 takes $q = 1$ and all directions at once.

12.2.3 Shadows

One inequality gives the extent of an ellipsoid in every direction.

Lemma 12.2.3 — Extent of an ellipsoid

Let $\mathbf{W}$ be a $q \times q$ positive definite matrix and $c > 0$. For every $\mathbf{a} \in \mathbb{R}^q$,

$$\max\{\mathbf{a}^\top \mathbf{u} : \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq c\} = \sqrt{c \mathbf{a}^\top \mathbf{W} \mathbf{a}},$$

attained at $\mathbf{u} = \sqrt{c/(\mathbf{a}^\top \mathbf{W} \mathbf{a})} \mathbf{W} \mathbf{a}$ when $\mathbf{a} \neq \mathbf{0}$. Consequently $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq c$ iff $|\mathbf{a}^\top \mathbf{u}| \leq \sqrt{c \mathbf{a}^\top \mathbf{W} \mathbf{a}}$ for every $\mathbf{a} \in \mathbb{R}^q$.

Proof. The bound is Corollary 1.7.8(c) with $\mathbf{B} = \mathbf{W}^{-1}$; directly, write $\mathbf{a}^\top \mathbf{u} = (\mathbf{W}^{1/2} \mathbf{a})^\top (\mathbf{W}^{-1/2} \mathbf{u})$. By the Cauchy–Schwarz inequality, $(\mathbf{a}^\top \mathbf{u})^2 \leq (\mathbf{a}^\top \mathbf{W} \mathbf{a})(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}) \leq c \mathbf{a}^\top \mathbf{W} \mathbf{a}$, with equality when $\mathbf{W}^{-1/2} \mathbf{u}$ is a nonnegative multiple of $\mathbf{W}^{1/2} \mathbf{a}$ and $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} = c$, which is the stated $\mathbf{u}$. For the last statement, “only if” follows from the maximum applied to $\mathbf{a}$ and $-\mathbf{a}$. For “if”, take $\mathbf{a} = \mathbf{W}^{-1} \mathbf{u}$: then $(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u})^2 \leq c \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}$, so $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq c$. ■

Corollary 12.2.4 — Shadows of the confidence ellipsoid

For every $\mathbf{a} \in \mathbb{R}^q$, the projection of $E(\mathbf{Y})$ onto the line spanned by $\mathbf{a}$ is the interval

$$\mathbf{a}^\top \hat{\boldsymbol{\phi}} \pm \sqrt{q F_\alpha(q, n-r)} \text{se}(\mathbf{a}^\top \hat{\boldsymbol{\phi}}), \quad \text{se}(\mathbf{a}^\top \hat{\boldsymbol{\phi}}) = s \sqrt{\mathbf{a}^\top \mathbf{W} \mathbf{a}},$$

and $E(\mathbf{Y})$ is the intersection of the slabs $\{\boldsymbol{\phi}_0 : |\mathbf{a}^\top \hat{\boldsymbol{\phi}} - \mathbf{a}^\top \boldsymbol{\phi}_0| \leq \sqrt{q F_\alpha} \text{se}(\mathbf{a}^\top \hat{\boldsymbol{\phi}})\}$ over all $\mathbf{a}$. Hence, with probability exactly $1 - \alpha$, every one of these intervals contains $\mathbf{a}^\top \mathbf{\Lambda}^\top \boldsymbol{\beta}$.

Proof. Apply Lemma 12.2.3 to $\mathbf{u} = \boldsymbol{\phi}_0 - \hat{\boldsymbol{\phi}}$ with $c = q s^2 F_\alpha$. The extreme values of $\mathbf{a}^\top \boldsymbol{\phi}_0$ over $E(\mathbf{Y})$ are $\mathbf{a}^\top \hat{\boldsymbol{\phi}} \pm \sqrt{c \mathbf{a}^\top \mathbf{W} \mathbf{a}}$, and the set of values in between is attained because $E(\mathbf{Y})$ is convex. The intersection statement is the last part of the lemma. The final claim holds because the event “all intervals cover” is the event $\boldsymbol{\phi} \in E(\mathbf{Y})$.

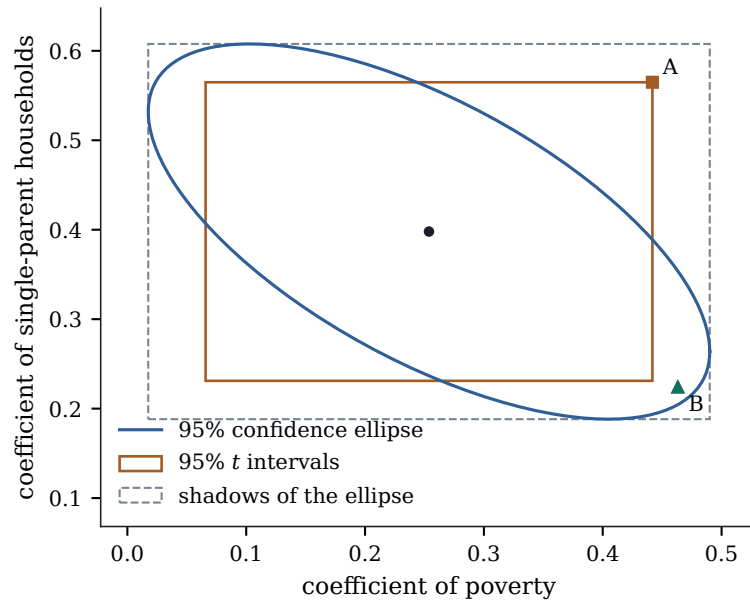


Figure 12.2.1: The 95% confidence ellipse for the poverty and single-parent coefficients, the box of 95% t intervals, and the ellipse's shadows (dashed). A is inside the box but outside the ellipse; B the reverse.

```
Fq = stats.f.ppf(0.95, q, n - p)

def F_stat(d):
    """F statistic for the hypothesis  $\Lambda^T \beta = d$ ."""
    u = phi_hat - d
    return u @ np.linalg.solve(W, u) / (q * s2)

def in_ellipse(d):
    return F_stat(d) <= Fq

corr = W[0, 1] / np.sqrt(W[0, 0] * W[1, 1])
print(f"estimates {phi_hat}, correlation of the estimates {corr:.3f}")
print(f"F({q}, {n - p}) 95% point {Fq:.3f}; F statistic for (0, 0): {F_stat(np.zeros(2)):.2f}")
```

[Run this code online](#)

```
tq = stats.t.ppf(0.975, n - p)
se = np.sqrt(s2 * np.diag(W))
t_box = np.column_stack([phi_hat - tq * se, phi_hat + tq * se])          # one-at-
    a-time t intervals
k = np.sqrt(q * Fq)
shadow = np.column_stack([phi_hat - k * se, phi_hat + k * se])        #
    projections of the ellipse
print("t intervals:\n", t_box.round(3))
print("shadows of the ellipse:\n", shadow.round(3), f"\nwider by the factor {k / tq:.3f}")
```

[Run this code online](#)

12.2.4 Two cautions

The ellipsoid is for a question fixed in advance. If the coefficients are chosen because they look interesting, the ball of [Proposition 12.2.2](#) or the methods of [Chapter 13](#) are the honest tools.

Intervals after a test. If intervals are reported only when a preliminary test of $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$ rejects, their coverage *given* the rejection is no longer $1 - \alpha$, and can be larger or smaller (Olshen 1973).

12.2.5 Exercises

12.2.5.1 A. Check your understanding

Exercise A1.

Show that for $q = 1$ the ellipsoid [Eq. 12.2.1](#) is the t interval [Eq. 12.1.1](#). Which fact about the $F(1, \nu)$ distribution does this use?

Solution. With $q = 1$, $\mathbf{W} = w$ is a positive number and the region is $(\hat{\phi} - \phi_0)^2 \leq s^2 w F_\alpha(1, n - r)$. If $T \sim t(\nu)$, then $T^2 \sim F(1, \nu)$ (Definition 4.1.3 and the remark after it), so $\Pr\{T^2 \leq t_{\nu, \alpha/2}^2\} = \Pr\{|T| \leq t_{\nu, \alpha/2}\} = 1 - \alpha$ and $F_\alpha(1, \nu) = t_{\nu, \alpha/2}^2$. The region is $|\hat{\phi} - \phi_0| \leq t_{\nu, \alpha/2} s \sqrt{w}$.

Exercise A2.

For the straight line $E Y_i = \beta_0 + \beta_1 x_i$, write the 95% confidence ellipse for (β_0, β_1) in terms of n , $\bar{x}$ and $\sum x_i^2$. Show that in the centred parameterization $E Y_i = \gamma_0 + \beta_1(x_i - \bar{x})$ the axes of the ellipse for (γ_0, β_1) are parallel to the coordinate axes. When is it a circle?

Solution. $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum x_i^2 \end{pmatrix}$, and the region is $n(\hat{\beta}_0 - b_0)^2 + 2n\bar{x}(\hat{\beta}_0 - b_0)(\hat{\beta}_1 - b_1) + \sum x_i^2(\hat{\beta}_1 - b_1)^2 \leq 2s^2 F_\alpha(2, n - 2)$. In the centred parameterization the cross-product matrix is $\text{diag}(n, S_{xx})$, so the quadratic form has no cross term, and the ellipse has axes along the coordinates with half-lengths $\sqrt{2s^2 F_\alpha/n}$ and $\sqrt{2s^2 F_\alpha/S_{xx}}$. It is a circle iff $S_{xx} = n$, that is, iff the x_i have mean square deviation one.

12.2.5.2 B. Practice

Exercise B1.

Take $q = 2$ and let ρ be the correlation of $\hat{\phi}_1$ and $\hat{\phi}_2$. Show that the area of the ellipse divided by the area of the box of the two t intervals is

$$\frac{\pi F_\alpha(2, n - r)}{2 t_{n-r, \alpha/2}^2} \sqrt{1 - \rho^2},$$

so that it depends on the design only through ρ . Check it against [Example \(A joint](#)

region for two coefficients), where the areas are 0.1197 and 0.1255 at $\rho = -0.639$. For which $|\rho|$ do the two regions have equal area there, and what happens to the ratio as the two regressors become collinear?

Solution. By Theorem 12.2.1(b) the ellipse has area $\pi \cdot 2F_\alpha s^2 \det(\mathbf{W})^{1/2}$, and writing ω_j for the diagonal entries of $\mathbf{W}$, $\det \mathbf{W} = \omega_1 \omega_2 (1 - \rho^2)$. The box has sides $2ts\sqrt{\omega_j}$, so area $4t^2 s^2 \sqrt{\omega_1 \omega_2}$; divide. In the example $F_{0.05}(2, 46) = 3.200$ and $t = 2.013$ give 1.240 at $\rho = 0$, and $1.240\sqrt{1 - \rho^2} = 0.954$ at $\rho = -0.639$, which is $0.1197/0.1255$. The areas agree when $\sqrt{1 - \rho^2} = 1/1.240$, that is $|\rho| = 0.592$. With uncorrelated estimates the ellipse is the larger region; as the regressors become collinear, $|\rho| \rightarrow 1$, the ratio tends to 0 and the ellipse becomes a thin needle across a large box, which the joint region mostly excludes.

Exercise B2.

In Example (A joint region for two coefficients), the hypothesis that one point of poverty and one point of single parenthood have the same coefficient is the line $\phi_1 = \phi_2$. Using Lemma 12.2.3, show that this line meets the ellipse iff $|\hat{\phi}_1 - \hat{\phi}_2| \leq \sqrt{2F_\alpha(2, n - r)} \text{se}(\hat{\phi}_1 - \hat{\phi}_2)$. Compare this with the t test of $\phi_1 = \phi_2$. Which is appropriate if the hypothesis was formulated before seeing the data?

Solution. The line is $\{\phi_0 : \mathbf{a}^\top \phi_0 = 0\}$ with $\mathbf{a} = (1, -1)^\top$. By Corollary 12.2.4, the values of $\mathbf{a}^\top \phi_0$ over the ellipse fill the interval $\mathbf{a}^\top \hat{\phi} \pm \sqrt{2F_\alpha} s \sqrt{\mathbf{a}^\top \mathbf{W} \mathbf{a}}$, so the line meets the ellipse iff this interval contains 0. The t test uses the smaller multiplier $t_{n-r, \alpha/2}$. It is the right test for a hypothesis fixed in advance. The ellipse criterion is the right one for a contrast chosen after looking at the estimates, because it holds simultaneously for all $\mathbf{a}$.

Exercise B3.

Let the columns of $\mathbf{\Lambda}$ lie in $\mathcal{C}(\mathbf{X}^\top)$ but be linearly dependent, with $\text{rank}(\mathbf{\Lambda}) = k < q$. Show that $\mathbf{W} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ is singular, that $\hat{\phi} - \phi \in \mathcal{C}(\mathbf{W})$ with probability one, and that

$$\begin{aligned} \{ \phi_0 \in \hat{\phi} + \mathcal{C}(\mathbf{W}) : (\hat{\phi} - \phi_0)^\top \mathbf{W}^{-} (\hat{\phi} - \phi_0) \\ \leq k s^2 F_\alpha(k, n - r) \} \end{aligned}$$

is an exact $1 - \alpha$ confidence set, a k -dimensional ellipsoid inside a k -dimensional affine subspace of $\mathbb{R}^q$.

Solution. $\mathbf{W} = (\mathbf{M}\mathbf{T})^\top (\mathbf{M}\mathbf{T})$ has rank $\text{rank}(\mathbf{M}\mathbf{T}) = \text{rank}(\mathbf{\Lambda}) = k$ (proof of Corollary 7.5.3). The vector $\hat{\phi} - \phi$ is $N_q(\mathbf{0}, \sigma^2 \mathbf{W})$, so it lies in $\mathcal{C}(\mathbf{W})$ almost surely (Theorem 2.2.3). By Corollary 7.5.3 the quadratic form over $k s^2$ is $F(k, n - r)$ and does not depend on the generalized inverse. So the displayed set covers ϕ with probability $1 - \alpha$. Restricted to the affine subspace, write $\mathbf{W} = \mathbf{U}\mathbf{D}\mathbf{U}^\top$ with $\mathbf{U}$ ($q \times k$) orthonormal and $\mathbf{D}$ positive diagonal. For $\phi_0 - \hat{\phi} = \mathbf{U}\mathbf{c}$, the form is $\mathbf{c}^\top \mathbf{D}^{-1} \mathbf{c}$, an ellipsoid in the coordinates $\mathbf{c}$.

12.2.5.3 C. Going deeper

Exercise C1.

Let $\mathbf{X}$ have full column rank. Show that the confidence ellipsoid for β is a region of the form $\{\mathbf{b} : \ell_p(\mathbf{b}) \geq \ell_p(\hat{\beta}) - k\}$, where $\ell_p(\mathbf{b}) = \max_{\sigma^2} \ell(\mathbf{b}, \sigma^2)$ is the profile log-likelihood of [Chapter 7](#), and find k in terms of $F_\alpha(p, n-p)$. How does k compare with the value $\frac{1}{2}\chi_\alpha^2(p)$ suggested by the asymptotic theory of likelihood ratios?

Solution. The profile log-likelihood is $\ell_p(\mathbf{b}) = \text{const} - \frac{n}{2} \log \|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2$, and $\|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE} + (\hat{\beta} - \mathbf{b})^\top \mathbf{X}^\top \mathbf{X} (\hat{\beta} - \mathbf{b})$ by Eq. 6.3.1. So $\ell_p(\hat{\beta}) - \ell_p(\mathbf{b}) = \frac{n}{2} \log\{1 + pF_{\mathbf{b}}/(n-p)\}$, where $F_{\mathbf{b}}$ is the statistic of [Theorem 12.2.1\(c\)](#). This is increasing in $F_{\mathbf{b}}$, so the ellipsoid is the region with $k = \frac{n}{2} \log\{1 + pF_\alpha(p, n-p)/(n-p)\}$. As $n \rightarrow \infty$, $pF_\alpha(p, n-p) \rightarrow \chi_\alpha^2(p)$ and $k \rightarrow \frac{1}{2}\chi_\alpha^2(p)$. For finite n , k is larger, and the exact region is larger than the asymptotic likelihood region.